

• Edit Distance / Sequence Alignment

ASSIGNMENT
ALGORITHMS

edits ↗ gaps
↘ mismatches

e.g. occurrence and occurrence
o-currence | o-curr-ance
occurrence | occurre-nce
= 1 gap, 1 mismatch | = 3 gaps, no mismatch

Q. Min. entries to be changed to get the other word.

a₁ a₂ ... a_m
b₁ b₂ ... b_n

A A R O N
A R O N x
4 edits

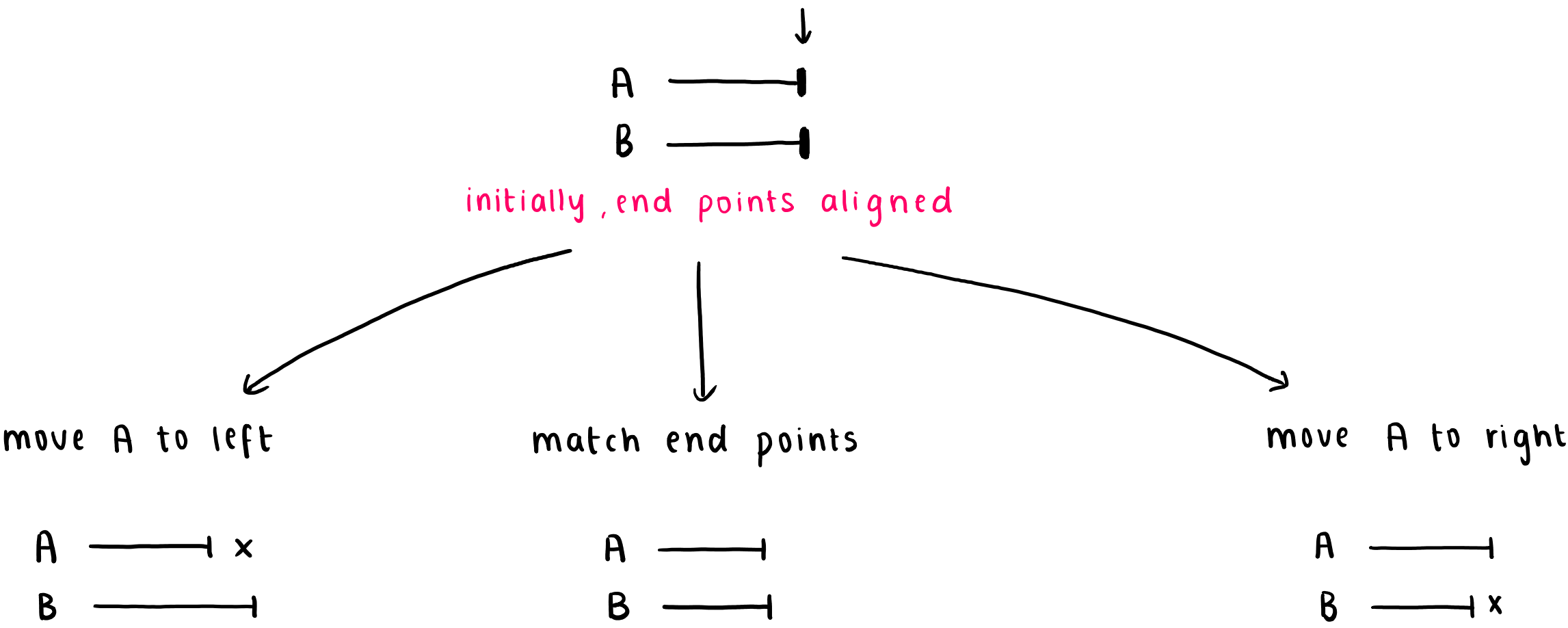
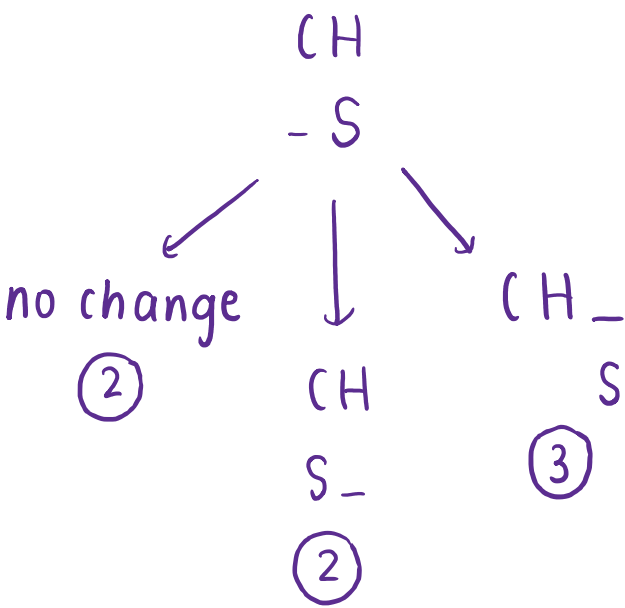
With alignment: A A R O N
A R O N
1 edit

C H A N D R A
S _ A N D R A
2 edits

* Optimal Sequence Alignment :

A[1,m]
B[1,n]

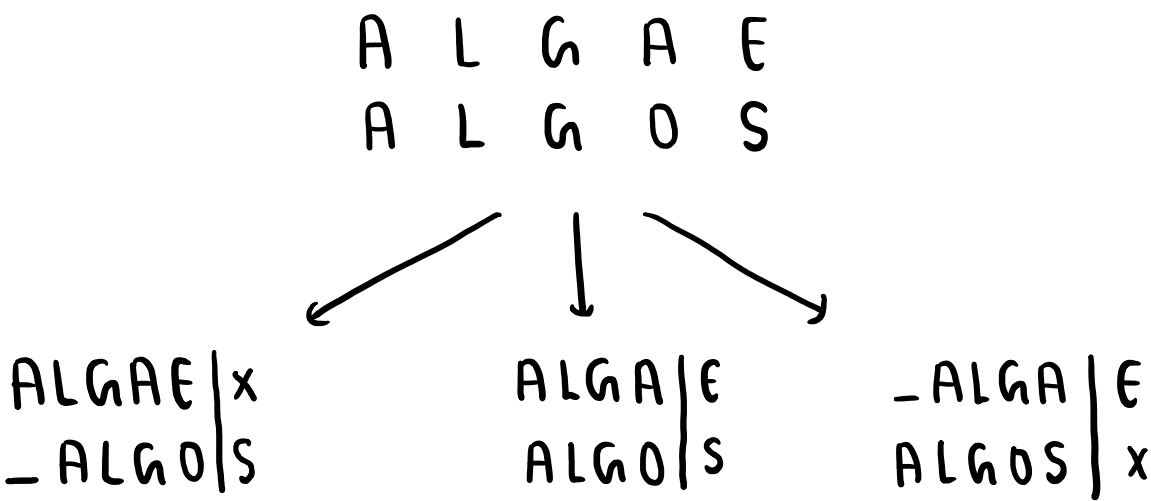
Base case: one letter α
β
α = β ⇒ edit = 0
α ≠ β ⇒ edit = 1



$\Delta(A[1,m], B[1,n-1]) + 1$ $\Delta(A[1,m-1], B[1,n-1]) + b$ $\Delta(A[1,m-1], B[1,n]) + 1$
where $b = \begin{cases} 1, & \text{if } A_m \neq B_n \\ 0, & \text{otherwise} \end{cases}$

∴ Working right to left :

$\Delta(A[1,m], B[1,n]) = \min \begin{cases} \Delta(A[1,m], B[1,n-1]) + 1 \\ \Delta(A[1,m-1], B[1,n]) + 1 \\ \Delta(A[1,m-1], B[1,n-1]) + b \end{cases}$



Working left to right:

